Five ways to mismeasure a post-quantum TLS 1.3 handshake on a narrowband link

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ABSTRACT

Post-quantum migration is increasingly argued from measurements: run the new primitives on a real stack and report what the handshake costs. On a narrowband command-and-control link such numbers are hard to win and easy to lose, because the failure modes of the measurement itself produce values that look exactly like results. We document five ways our own first pass was wrong, each of which produced a plausible number: a token bucket that credited the same wait twice and made the link run 53 % fast; an initial burst that gave the first 1.17 s away free; a server status page that contributed 72 % of what was being counted; session tickets that fell inside the measurement window on some runs and outside it on others, moving the byte count by 12 %; and a certificate-authority option that copied a 5.7-kB certificate into every session ticket, inflating the reported handshake by 54.7 %. Each pitfall is stated with its symptom, its magnitude, and the check that caught it. The checks compose into a protocol — record-layer attribution, a byte-exact cross-check against the standards, per-direction totals, and a canonical flag set — that we recommend to anyone measuring TLS on a constrained link. Applied to a mutual-authentication ML-KEM-768+ML-DSA-65 handshake the protocol yields 20,840 bytes, of which 87.9 % is identity material: a number that survives all five checks.

Keywords: post-quantum cryptography, TLS 1.3, performance measurement,
constrained links, unmanned aerial vehicles, measurement pitfalls

1. INTRODUCTION

Post-quantum cryptography has moved from standards to deployment: FIPS 203 and FIPS 204 specify ML-KEM and ML-DSA[1,2], FIPS 205 adds SLH-DSA[3], X.509 identifiers exist for both[4,5], and mainstream libraries ship implementations[6]. The migration is forced by Shor's algorithm[7] and by the harvest-now-decrypt-later argument that Mosca's inequality quantifies[8]; the process behind the standards is documented in[9] and surveyed in[10,11]. A growing literature measures what the primitives cost on the platforms that will have to run them, from Cortex-M4 cycle counts[12,13] to TLS handshake timings on desktop stacks[14,15], and the first UAV studies are appearing[16,17].

Measured numbers in this area travel far. A byte count that enters a comparison table is quoted, re-quoted, and eventually becomes the basis of a migration decision, yet the measurement itself is rarely scrutinised. The errors a measurement harness makes are not random; they are *plausible*. A status page is traffic on the socket, a session ticket is part of the handshake, an initial burst is what sockets do — each produced a credible number in our own first pass, and nothing about the number revealed the error.

This paper is an account of five such failures, in the order we met them, with the magnitude of each and the check that caught it. The setting is a measurement that was easy to get wrong in several independent ways at once: a mutual-authentication TLS 1.3 handshake carrying ML-KEM-768 and ML-DSA-65[1,2], measured over an emulated 9.6-kbps, 250-ms beyond-line-of-sight command-and-control link of the kind surveyed in[18] and specified in[19], carrying MAVLink telemetry[20], where the whole handshake takes eleven seconds and every artefact is worth seconds. The contributions are the failure catalogue (Section 3), a detection protocol assembled from the checks that caught them (Section 4), and the corrected number that survives the protocol (Section 5). A fuller account of the underlying study — rate, delay and loss sweeps, and a comparison of remediation options — is in preparation; this paper deliberately reports only what is needed to make the measurement methodology checkable. The harness and all raw measurements are public[21].

2. TESTBED AND MEASUREMENT CONVENTION

Stack and configuration

All measurements use OpenSSL 3.5.7[6] with TLS 1.3 as specified in RFC 8446[22]. The measured configuration is a mutual-authentication handshake in which both peers present an ML-DSA-65 certificate[2] and the key exchange is the pure ML-KEM-768 group[1]. The other standardised signature family, SLH-DSA, is not measured here: its smallest parameter set carries 7,856-byte signatures[3], which a 9.6-kbps link cannot absorb in any useful number of handshakes. Research libraries such as liboqs offer a wider algorithm set but no record layer[23], and the record layer is what this paper measures. A classical comparison configuration uses X25519 with ECDSA-P256; it measures 1,915 bytes and 1.99 s against the post-quantum 20,840 bytes and 11.27 s, a factor of 5.66 in time on the link used throughout. Certificates are self-signed at chain depth one, and session tickets are disabled on both sides by default — for a reason that turns out to be pitfall 5.

Link emulation

The client and server run on one host and communicate through a userspace TCP proxy that applies a token bucket and a fixed one-way propagation delay per forwarded chunk in each direction: 9.6 kbps per direction and 250 ms, the parameters of a narrowband beyond-line-of-sight radio. DTLS 1.3 is the transport such a link would use in practice, but this stack implements DTLS 1.2 only[24,25], whose cipher-suite names encode the authentication algorithm and which has no ML-KEM group; TLS over TCP is therefore what can be measured, and the framing difference is a bound rather than a measurement. The throttle was validated against a known transfer: pushing 12,000 bytes through the 9.6-kbps setting yields 9.57 kbps, within 0.3 % of target. Two earlier versions of the throttle were wrong in ways that were invisible in the handshake numbers themselves; both are pitfalls 1 and 2 below.

Byte accounting convention

Bytes are counted by a record-layer parser inside the proxy, not by a socket counter: it records every TLS record with its type and length and attributes the two plaintext hellos to message types, so everything after the ServerHello is accounted by record. The client's one-byte liveness probe occupies a 23-byte record, which every total here includes; the convention is stated so it can be undone by hand. Byte counts include 5-byte TLS record headers, key-exchange payloads do not.

3. FIVE PITFALLS

Table 1 summarises the five failure modes. None of them is exotic, and that is the point: each produced a number we were ready to report.

Table 1. The five measurement pitfalls. The middle columns give the effect each had on the reported number before it was found, and the last column the change that removed it. No number in this paper predates the fixes.

#	Failure mode	Effect on the reported number	Correction
1	Token bucket credited the same wait twice	link ran at 14.73 instead of 9.60 kbps, 53 % fast	allow the balance to go negative
2	Free initial burst of 1,400 bytes	1.17 s uncharged at 9.6 kbps	burst set to zero
3	`-www` status page counted as handshake	about 5 kB of HTML against a 1.9 kB handshake, 72 % of the count	server run without `-www`
4	`NewSessionTicket` inside or outside the window	12 % byte variation between runs	tickets disabled on both sides
5	`-CAfile` certificate copied into each ticket	20,840 → 32,236 bytes, +54.7 %	`-no_ticket` on the server as well

(1) The token bucket credited the same wait twice. The first throttle clamped the token balance to zero when a chunk exceeded the available credit, then refilled from elapsed time on the next call. The wait already served therefore re-entered the balance as fresh capacity, and the emulated link ran at 14.73 kbps against a 9.6-kbps target — 53 % fast. Every completion time in that version of the harness was compressed, and nothing about the compressed times looked wrong: they were smooth, monotone in bytes, and stable across runs. The fix is to let the balance go negative, so that the bucket owes capacity rather than forgetting the debt.

(2) The token bucket handed out a free burst. With a 1,400-byte initial burst, a short transfer received its first 1,400 bytes at no cost. At 9.6 kbps that is 1.17 s of airtime given away, which is not negligible when the measured handshake takes 11.27 s: a systematic 10 % discount on every transfer, paid to the first 1,400 bytes. The effect is not an artefact of our emulator alone: TCP opens a connection with a ten-segment initial window[26], so a real link hands out a burst of its own, and the choice here is to remove it and measure serialisation rather than to leave it unaccounted. The burst is now zero, which makes serialisation exact. The general lesson is that throttles must be validated against a known transfer, not trusted to be conservative.

(3) The server's status page was counted as handshake. An early version drove `s_server` with `-www`, whose HTML status page is about 5 kB and was counted in the byte total. On the classical configuration this was more than half the reported figure: about 5 kB against a 1.9-kB handshake, 72 % of what was being counted. The error survived a plausibility check that should have killed it — 7 kB felt large for a handshake — because the post-quantum configurations in the same batch were larger still. The server now runs without `-www`.

(4) Session tickets fell inside or outside the window. `NewSessionTicket` arrives after the handshake completes, so whether it fell inside the measurement window depended on link speed and scheduling. The same configuration varied by 12 % in byte count between runs. This is the failure mode that record-level accounting is for: a total counter cannot distinguish a handshake that carries tickets from one that does not, and a window is a policy decision that must be stated. Tickets are now disabled on both sides, which makes every run byte-identical.

(5) The certificate-authority certificate was embedded in every ticket. This is the one that mattered most, because it inflated the result we most wanted to be true. With client authentication enabled, `s_server` uses its `-CAfile` both to validate the client certificate and to build the chain it sends, and with tickets enabled the CA certificate reappeared inside each `NewSessionTicket`: the two post-handshake records grew from 79 bytes each — what a ticket should cost — to 5,775 bytes each, 5,696 bytes larger, the size of the certificate plus ticket framing. The server-to-client direction grew from 10,468 to 21,860 bytes, and the handshake total from 20,840 to 32,236 bytes, an inflation of 54.7 % (Figure 1). Client-side `-no_ticket` does not suppress it, because the ticket is emitted by the server after the handshake; the server must be given the flag as well. Two details make the account checkable to the byte. The client's flight gains four bytes at the same time, because with tickets enabled the client offers the empty `session_ticket` extension, which `-no_ticket` suppresses, growing its `ClientHello` from 1,357 to 1,361 bytes; the two directions therefore grow by 11,392 and 4 bytes, and the 11,396-byte growth of the total is their sum. And a 32-kB handshake is not obviously wrong — published post-quantum figures are in that range — so what caught this was not intuition but attribution: the 5,775-byte records were not explicable from any field of the handshake, and a record that cannot be attributed has to be investigated rather than averaged in.

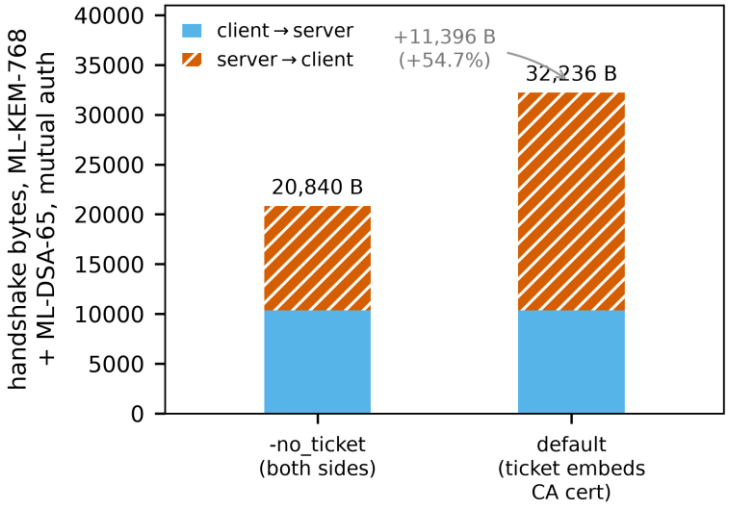


Fig. 1. Pitfall 5. Both bars are the same mutual-authentication ML-KEM-768+ML-DSA-65 handshake. In the left bar the server was started with -no_ticket and the two post-handshake records are 79 bytes each; in the right bar that flag was omitted and the -CAfile certificate is embedded in each ticket. Reporting the right-hand bar as the cost of post-quantum authentication would have overstated it by 54.7%.

4. A DETECTION PROTOCOL

Five one-off catches are an anecdote; the useful output is the check set that caught them, which generalises beyond this testbed. We recommend four checks for any TLS measurement on a constrained link.

Attribute by record, not by socket. Count bytes at the record layer with a parser that names every record, not with a counter on the socket or the process. Pitfalls 3, 4 and 5 were all invisible to a total counter and trivial to a parser: a status page is application data, a ticket is a named post-handshake record, and an oversized ticket is a record whose length no handshake field explains. The parser needs no decryption — TLS 1.3 record boundaries and lengths are visible without keys, and the two hellos are plaintext.

Cross-check one record against the standards to the byte. At least one measured record should be reconstructed from first principles. The 5,549-byte `Certificate` record is 35 bytes of TLS framing (5 record header, 16 AEAD tag, 4 handshake header, 1 content type, 1 context length, 3 list length, 3 certificate length, 2 extensions length) plus 5,514 bytes of DER, which is in turn 1,952 of ML-DSA-65 public key, 3,309 of signature and 253 of X.509 framing[2,4]. Table 2 gives the full per-message reconstruction. A check of this kind is worth more than repetition: repetition bounds noise, while a byte-exact reconstruction detects systematic error, and it was exactly this check — not a statistic — that exposed pitfalls 3 and 5.

Table 2. Per-message reconstruction of the corrected handshake. Bytes in the second column are read off the wire by the record parser; bytes in the third column follow from FIPS 203/204 and X.509 framing, so the two columns check each other. Framing rows hold the records that carry no handshake content: ChangeCipherSpec, the 23-byte liveness probe (client), the two tickets (server, 79 bytes each here), and close_notify.

Message	Bytes	Contents
client → server (total 10,372)		
`ClientHello`	1,357	key share 1,192
`Certificate`	5,549	DER 5,514 + 35 framing
`CertificateVerify`	3,339	signature 3,309 + 30
`Finished`	74	
framing	53	CCS 6, probe 23, close 24
server → client (total 10,468)		
`ServerHello`	1,183	key share 1,096
`EncryptedExtensions`	28	
`CertificateRequest`	107	
`Certificate`	5,549	as above
`CertificateVerify`	3,339	as above
`Finished`	74	
framing	188	CCS 6, tickets 158, close 24
Total	20,840	

Replay with a canonical flag set. Tickets are disabled on both sides, the server runs without `-www`, the client sends a stated one-byte probe, and the throttle has no burst. The flag set is part of the method: half of the pitfalls above are impossible with the right flags, and a measurement that does not state its flags cannot be reproduced at the byte level at all, because byte counts are a deterministic function of the flags.

Report per-direction totals. The two flights of a handshake are 10,372 and 10,468 bytes, and artefacts do not respect their sum: an oversized ticket inflates only the server's flight, a status page only whichever side emits it. A single total hides the very asymmetry that makes an error visible. The 4-byte `session_ticket` extension of pitfall 5 would have been unfindable in a 32,236-byte total; in per-direction totals it is the only unexplained delta on the client flight.

Two further practices bound the residual error. The throttle is validated against a known transfer (within 0.3 % here), and the measurement is repeated: five runs give a standard deviation of 0.011 s, or 0.10 % of the mean. Repetition carries a caution of its own, though — in a batch of fifteen runs two stalled by 0.4–0.5 s with no injected loss, and while the means reproduced across batches within 1 %, the spreads did not. Repetition measures noise on the day; on a constrained link the tails are part of the measurement.

5. CASE STUDY: THE NUMBER THAT SURVIVES

Applying the protocol to the post-quantum handshake gives 20,840 bytes in 11.27 s at 9.6 kbps per direction and 250 ms one-way. Of those, 2,530 bytes (12.1 %) are the key exchange — both hellos and their record headers — and 18,310 (87.9 %) are identity material: two 5,549-byte certificates, two 3,339-byte `CertificateVerify` signatures, and the records that carry them. Each of the five pitfalls would have changed this headline. Pitfalls 1–3 distort the time axis; pitfalls 4 and 5 distort the byte axis, and pitfall 5 inflates the reported handshake by more than half while leaving the true handshake untouched. The 5.66× time ratio between the post-quantum and classical configurations — the number most likely to be quoted — becomes a byte ratio of 16.8× (32,236 against 1,915) and a time ratio of about 8.3× if the right-hand bar of Figure 1 is reported by mistake. None of the checks that caught these errors costs more than a few hundred lines of parsing code and the discipline to state a flag set; all are reproducible from the public repository[21].

6. RELATED WORK

Benchmarking of post-quantum primitives on constrained hardware is well developed[12,13], and the cost of the primitives inside TLS has been measured on desktop stacks[14,15,27], while KEMTLS shows how far the handshake itself can be restructured to remove the certificate signature[28]. That literature reports results; what it rarely reports is the failure modes of the measurement itself, which are exactly what a byte-level methodology paper can contribute. UAV-specific studies report a single capture[16] or a simulation[17], in neither case with a stated accounting convention. Our own record is no cleaner than the field's, which is the reason for this paper: for a while we attributed the oversized tickets of pitfall 5 to resumption rather than to `-CAfile`, and the plausibility argument behind that reading was no weaker than the one behind the correct reading. Every correction here is therefore stated with the observation that falsified the previous one — for pitfall 5, a 5,759-byte record with no field inside it large enough to explain the size.

7. CONCLUSION

Measured numbers about post-quantum handshakes are becoming the evidence base for migration decisions, and the measurement harness is the part of that evidence base that gets the least scrutiny. We document five failure modes that each produced a plausible number on a real TLS 1.3 stack, the largest inflating the result by 54.7 % and the subtlest moving it by four bytes, and we compress the checks that caught them into a four-part protocol: record-layer attribution, a byte-exact standards cross-check, a canonical flag set, and per-direction totals. The protocol costs little, needs no decryption, and converts a byte count from a claim into a derivation. We recommend it — and the reporting of negative results like these — as standard practice for constrained-link measurement.

8. ACKNOWLEDGMENTS

The measurement harness was developed with the assistance of AI coding tools under the authors' direction; all results were reviewed and reproduced by the authors, and the harness and raw measurements are publicly available.

9. REFERENCES

[1] National Institute of Standards and Technology, “FIPS 203: Module-lattice-based key-encapsulation mechanism standard,” tech. rep., U.S. Department of Commerce, 2024.

[2] National Institute of Standards and Technology, “FIPS 204: Module-lattice-based digital signature standard,” tech. rep., U.S. Department of Commerce, 2024.

[3] National Institute of Standards and Technology, “FIPS 205: Stateless hash-based digital signature standard,” tech. rep., U.S. Department of Commerce, 2024.

[4] J. Massimo, P. Kampanakis, and S. Salter, “Internet X.509 public key infrastructure: Algorithm identifiers for ML-DSA,” RFC 9881, 2025.

[5] J. Massimo, P. Kampanakis, and S. Salter, “Internet X.509 public key infrastructure: Algorithm identifiers for SLH-DSA,” RFC 9909, 2025.

[6] The OpenSSL Project, “OpenSSL 3.5: ML-KEM, ML-DSA and SLH-DSA support,” https://openssl-library.org/, 2025. Version 3.5.7 used for all measurements.

[7] P. W. Shor, “Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer,” SIAM Journal on Computing 26(5), pp. 1484–1509, 1997.

[8] M. Mosca, “Cybersecurity in an era with quantum computers: Will we be ready?,” IEEE Security & Privacy 16(5), pp. 38–41, 2018.

[9] L. Chen, S. Jordan, Y.-K. Liu, D. Moody, R. Peralta, R. Perlner, and D. Smith-Tone, “Report on post-quantum cryptography,” Tech. Rep. NISTIR 8105, National Institute of Standards and Technology, 2016.

[10] D. J. Bernstein and T. Lange, “Post-quantum cryptography,” Nature 549(7675), pp. 188–194, 2017.

[11] D. J. Bernstein, J. Buchmann, and E. Dahmen, eds., Post-Quantum Cryptography, Springer, Berlin, Germany, 2009.

[12] M. J. Kannwischer, J. Rijneveld, P. Schwabe, and K. Stoffelen, “pqm4: Testing and benchmarking NIST PQC on ARM Cortex-M4,” in Proc. 2nd NIST PQC Standardization Conference, 2019.

[13] L. Botros, M. J. Kannwischer, and P. Schwabe, “Memory-efficient high-speed implementation of Kyber on ARM Cortex-M4,” in Progress in Cryptology – AFRICACRYPT 2019, LNCS 11627, pp. 111–130, 2019.

[14] C. Paquin, D. Stebila, and G. Tamvada, “Benchmarking post-quantum cryptography in TLS,” in Post-Quantum Cryptography – PQCrypto 2020, LNCS 12100, pp. 72–91, Springer, 2020.

[15] D. Sikeridis, P. Kampanakis, and M. Devetsikiotis, “Post-quantum authentication in TLS 1.3: A performance study,” in Proc. 27th Annual Network and Distributed System Security Symposium (NDSS), The Internet Society, 2020.

[16] T. Xia, M. Wang, J. He, G. Yang, L. Fan, and G. Wei, “A quantum-resistant identity authentication and key agreement scheme for UAV networks based on Kyber algorithm,” Drones 8(8), 359, 2024.

[17] A. Hussien and A. A. Abdullah, “A post-quantum lightweight mutual authentication and encryption scheme for secure UAV communications,” Frontiers in Computer Science 8, 2026.

[18] Y. Zeng, R. Zhang, and T. J. Lim, “Wireless communications with unmanned aerial vehicles: Opportunities and challenges,” IEEE Communications Magazine 54(5), pp. 36–42, 2016.

[19] RTCA, Inc., “DO-362: Command and control (C2) data link minimum operational performance standards (MOPS) (terrestrial),” tech. rep., RTCA, Inc., Washington, DC, USA, 2016.

[20] MAVLink Development Team, “MAVLink developer guide,” https://mavlink.io/en/, 2024.

[21] X. Huang, Y. Han, S. Zhu, H. Wu, and Z. Liu, “Measurement package: post-quantum TLS 1.3 handshakes over an emulated narrowband command-and-control link,” https://github.com/huanghuayu12/pqc-uav-drones, 2026.

[22] E. Rescorla, “The transport layer security (TLS) protocol version 1.3,” RFC 8446, 2018.

[23] Open Quantum Safe Project, “liboqs: An open-source C library for quantum-safe cryptography,” https://openquantumsafe.org/, 2024.

[24] E. Rescorla, H. Tschofenig, and T. Fossati, “The datagram transport layer security (DTLS) protocol version 1.3,” RFC 9147, 2022.

[25] E. Rescorla and N. Modadugu, “Datagram transport layer security version 1.2,” RFC 6347, 2012.

[26] J. Chu, N. Dukkipati, Y. Cheng, and M. Mathis, “Increasing TCP’s initial window,” RFC 6928, 2013.

[27] D. Sikeridis, P. Kampanakis, and M. Devetsikiotis, “Assessing the overhead of post-quantum cryptography in TLS 1.3 and SSH,” in Proc. 16th International Conference on Emerging Networking Experiments and Technologies (CoNEXT), pp. 149–156, ACM, 2020.

[28] P. Schwabe, D. Stebila, and T. Wiggers, “Post-quantum TLS without handshake signatures,” in Proc. 2020 ACM SIGSAC Conference on Computer and Communications Security (CCS), pp. 1461–1480, ACM, 2020.